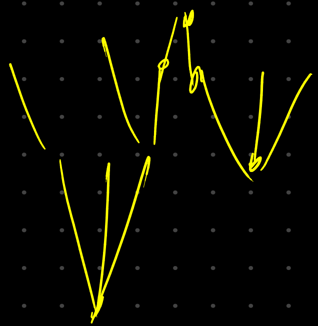
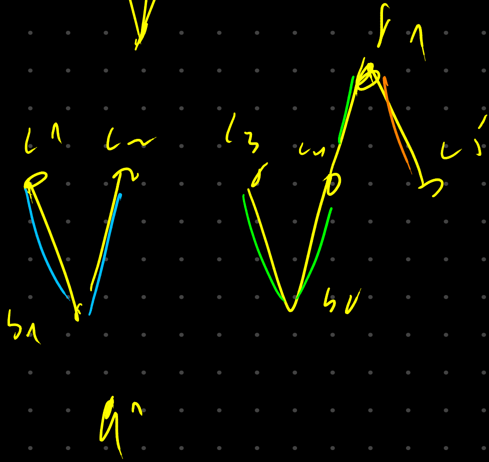
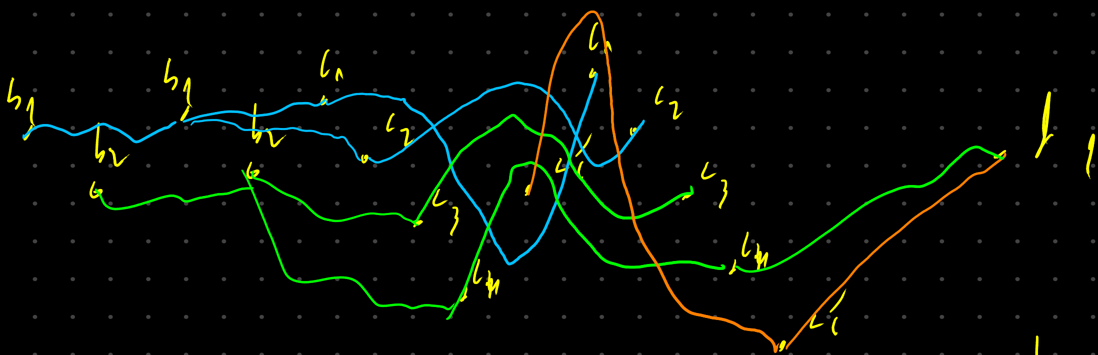
Plan:

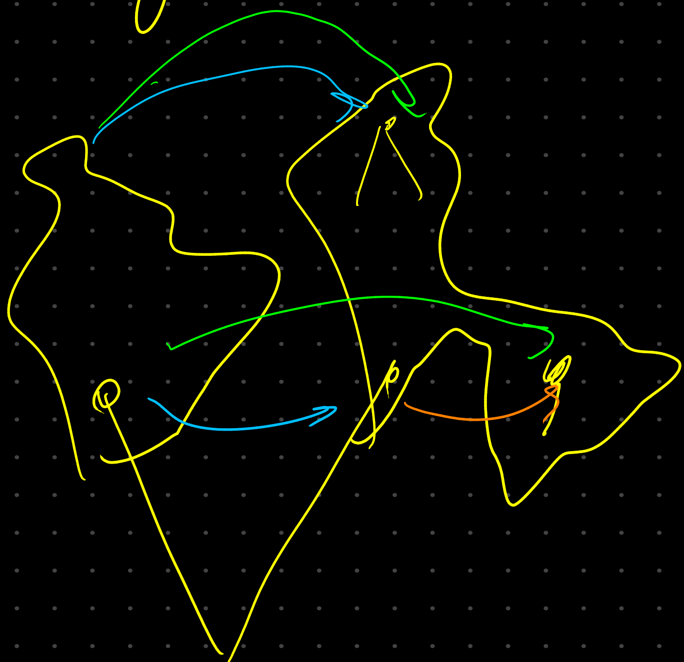
2 part of the thesis

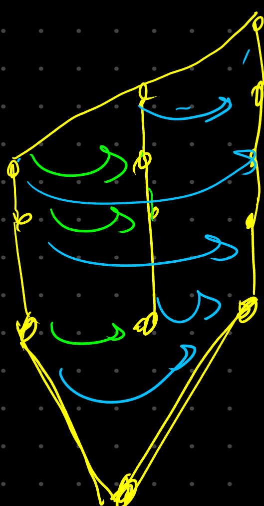






we can fix this by pushing the troublesome vertex out of the entangled node





if it's possible to do adj edges to warning then do it, if it's not then the vertex is not a problem

(if it's the same eq class then we are not in the case of

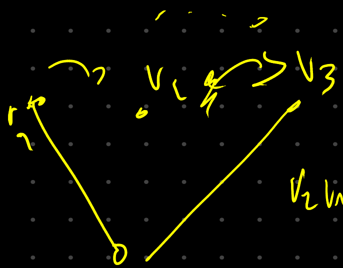
$\bar{Eq}(v_1,v_2) = \neg Eq(v_2 v_1)$

$\bar{Eq}(v_1,v_2) = Eq(v_1 v_2) \sqcup \neg \bar{Eq}(v_2 v_1)$

how to design the algorithm?

$\neg v_2 v_1,\ v_3 v_2$

$\bar{Eq}(v_1,v_2) = \{ v_1 v_2, \neg v_2 v_3 \}$



$v_2 v_1,\ v_2 v_3$

$\forall u \in V_{adj}$ $u v_2$ $\bar{E}(v_2,v_3) = \{ \neg v_1 v_2, v_2 v_3, v_2 v_1 \}$

$v_1 v_2$ $Eq(v_1 v_2) = v_1 v_2, v_3 v_2, \neg v_2 v_3$

- $v_2 v_3$ $\bar{Eq}(v_2 v_3) = v_2 v_3$, $\neg v_2 v_3$, $v_2 v_1$, $\neg v_1 v_2$
- $v_1 v_3 = v_1 v_3$

$v_1 v_2$ — $v_3 v_2$

$v_2 v_1$ — $v_2 v_3$

$$Eq(v\ v_2) = \bigsqcup_{v \in Adj} \bar{Eq}(v v_2)$$

$\bar{Eq}(v_1 v_2) =$

$Eq(v_3, v_3)$

$\bar{Eq}(v_1, v_3)$

$\bar{Eq}(v_1 v_2) = \bar{Eq}(v_3 v_2)$

if Eq

$V_{old}$

$\forall w \in V_{adj}$ $E(u,w)$

if $uw \in S$

else $V_f = V_0 \sqcup \{w u\}$

$S = \{v_1 v_2, v_1 v_3, v_1 v_3,$

$v_1 v_2$ $V_{opt} = \{v_1 v_2, v_1 v_3\}$ $v_2 v_3$

$v_1 v_3$

$v_1 v_3$ $V_{opt} = \{v_1 v_3, v_1 v_3, v_2 v_4, v_2 v_3, v_2 v_4, v_1 v_3\}$

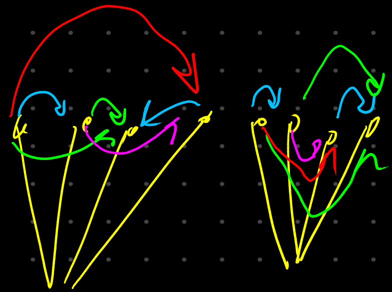
$v_2 v_3$

$v_2 v_4$

$v_3 v_3$

$f$: $V_0 := V_0 \sqcup \{uw\}$

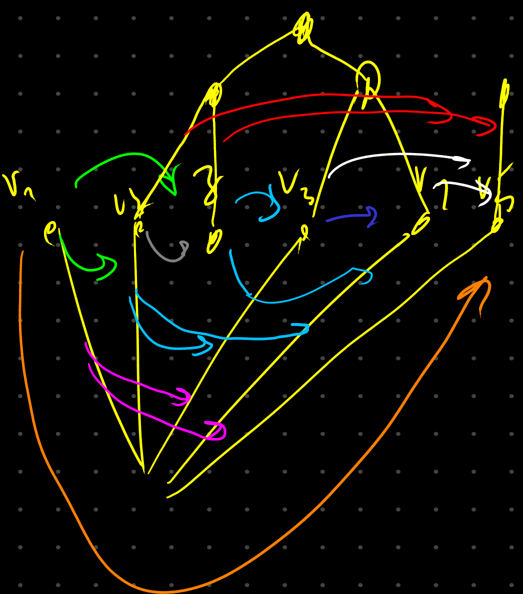
$S = S \sqcup E(wu)$



$$\forall uw \in V_0 \quad E_q(uw) = \bigsqcup_{u'w' \in V_0} E_q(u'w')$$

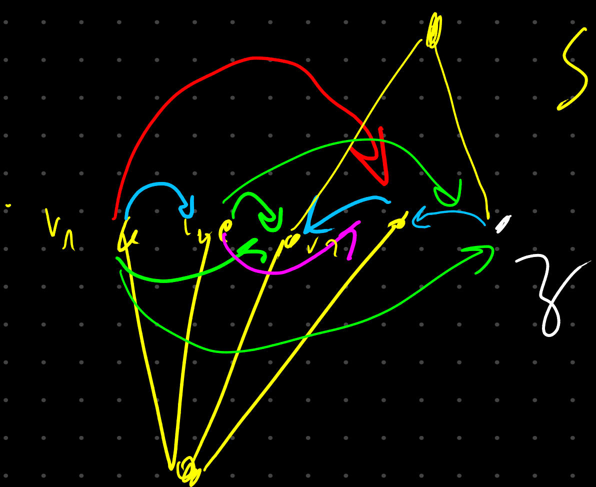
$$\forall wu \in V_0 \quad E_q(wu) = \{vu \,,\; uv \in E_q(uw)\}$$

We can do the same with the weak H-T case



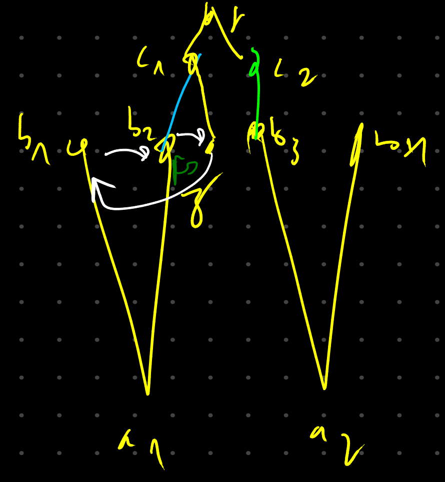

$$S = \{v_1v_2, v_1z, v_1v_3, v_1v_4, v_1v_5, v_2v_3, zv_3, v_2v_4, zv_4, v_2v_5, zv_5, v_3v_4, v_3v_5, v_4v_5\}$$

$$V_{ord} = \{\overbrace{v_1z, v_2z}^{v_1z}, zv_3, zv_4, zv_5\}$$

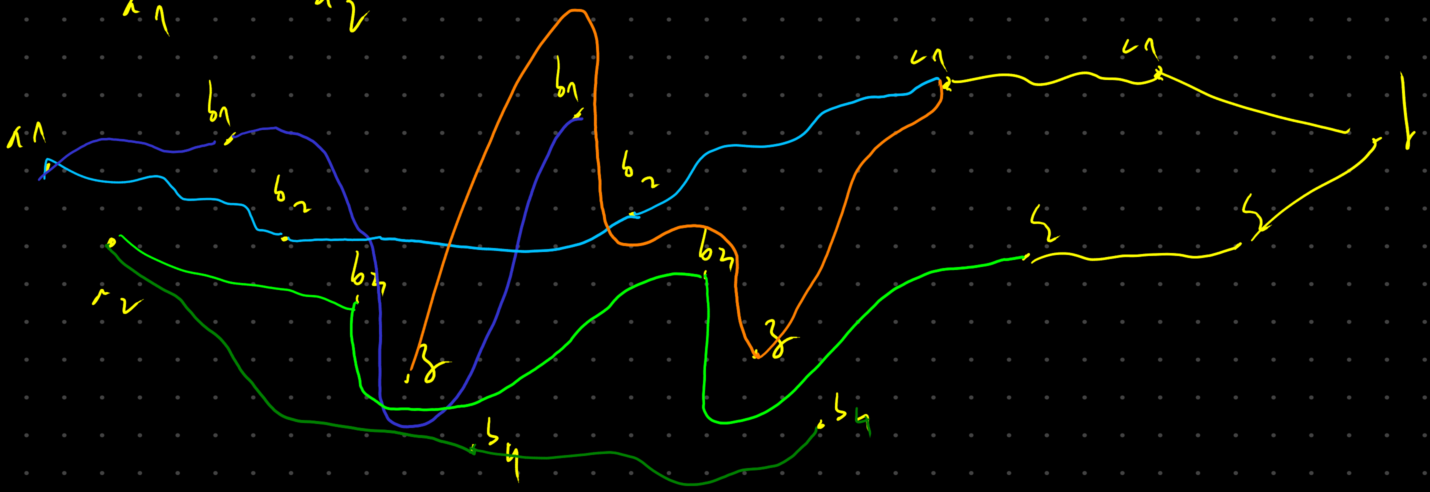



$$S = \{v_1v_2, v_4v_3, zv_4, v_1v_3, v_2v_3, v_2z, v_1z, v_1v_4, v_2v_4\}$$

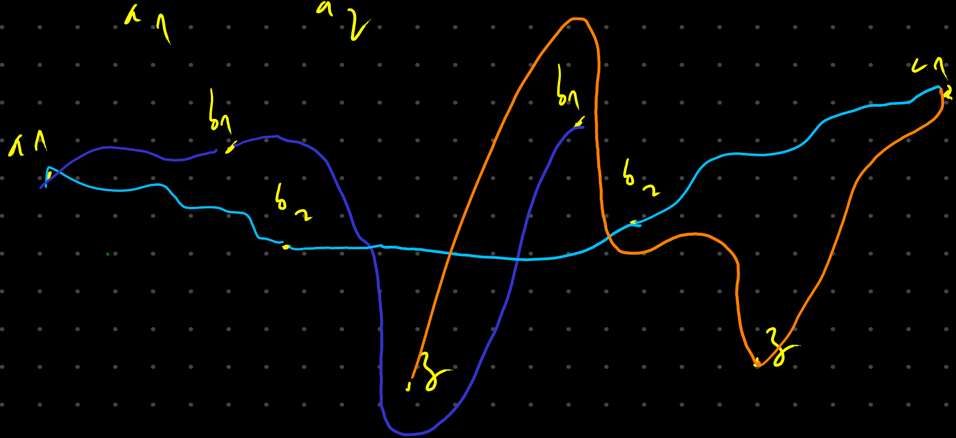
$$V_{ord} = \{v_1v_2, v_1v_3, v_1v_4, v_2v_3, v_2v_4, v_4v_3, v_1z, v_2z, v_3z, zv_4\}$$



$c_1$
$c_2$
$b_1$
$b_2$
$b_3$
$z$
$a_1$
$a_2$

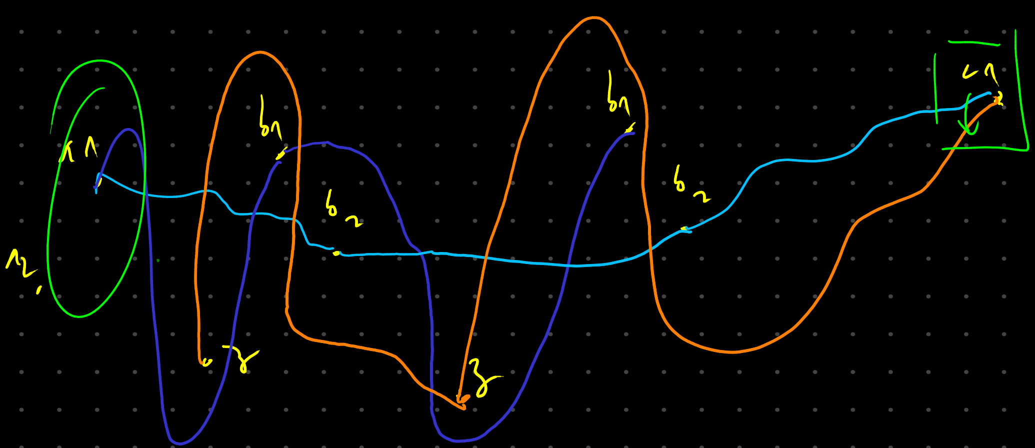



$a_1$
$b_1$
$b_2$
$c_1$
$z$
$a_2$
$b_4$

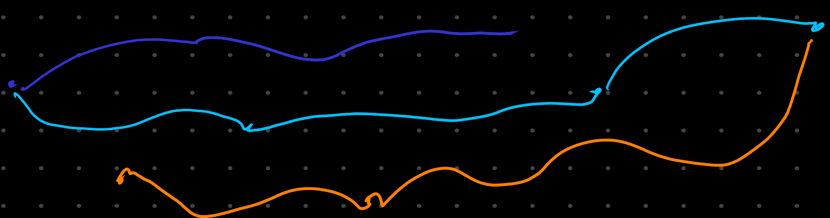



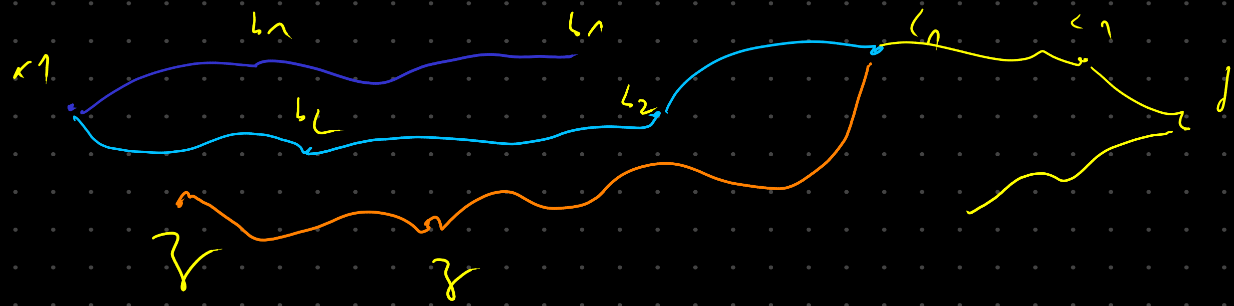
$a_1$
$a_2$
$b_1$
$b_2$
$c_1$
$z$

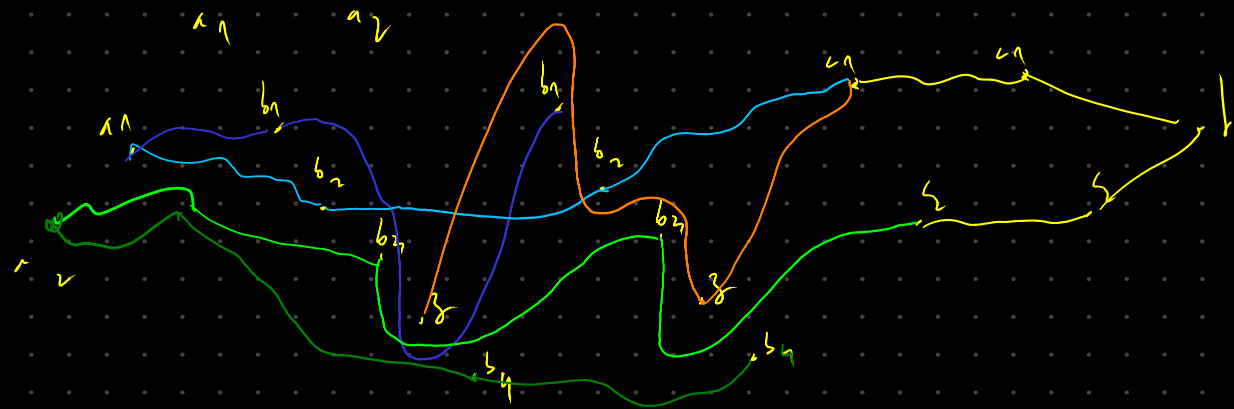

≡

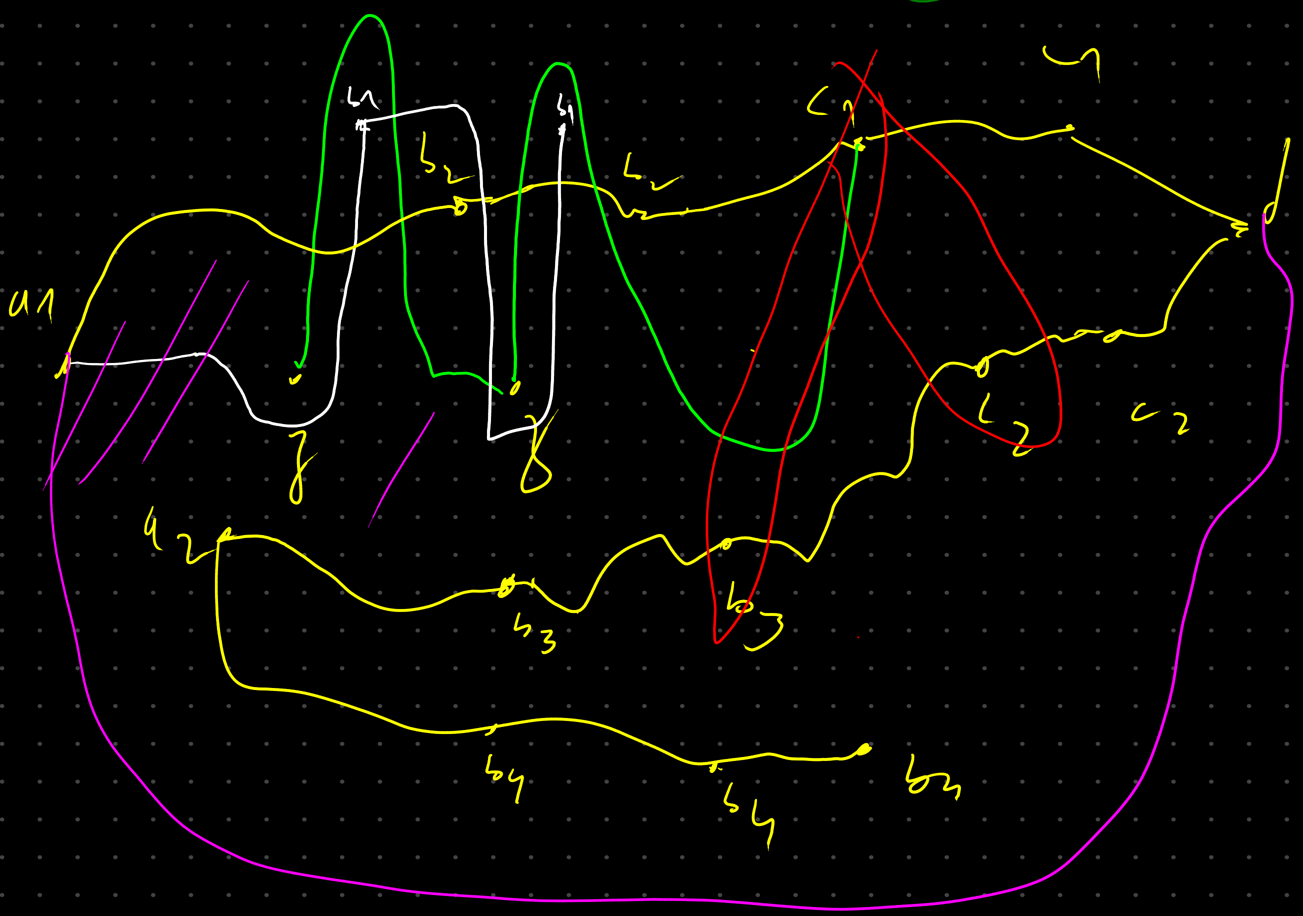


$a_1$
$a_2$
$b_1$
$b_2$
$c_1$
$z$



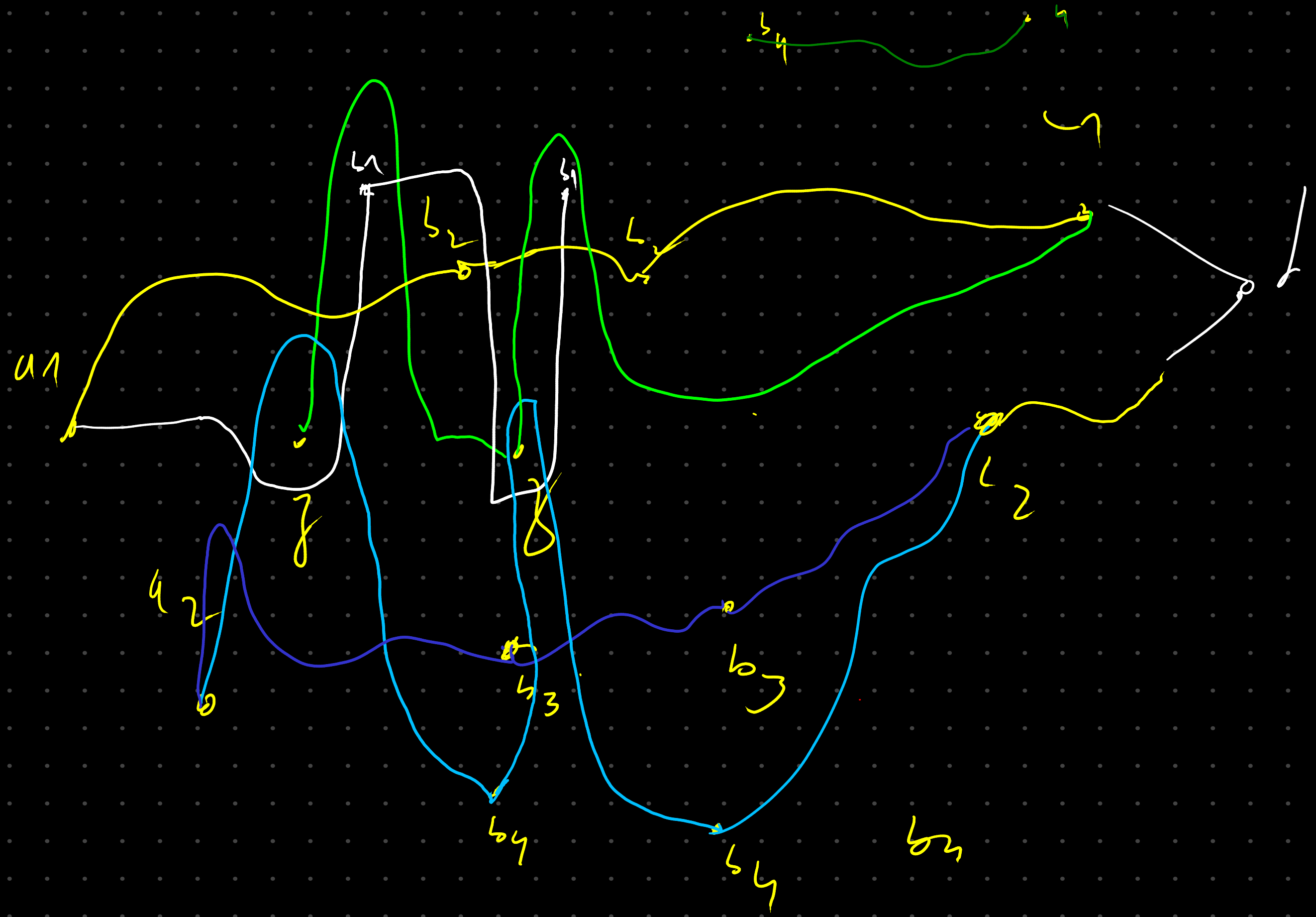



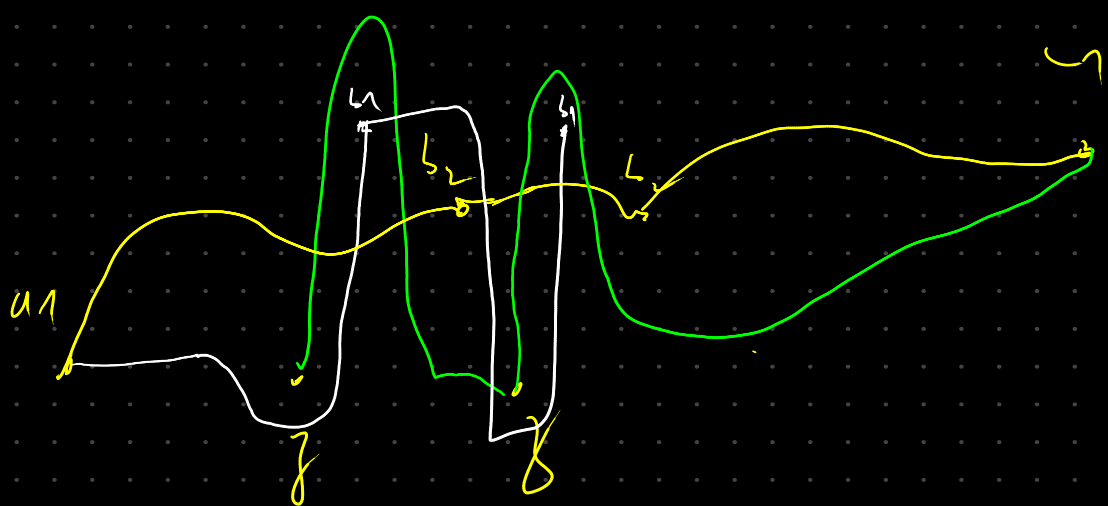

check relation of $c_1$

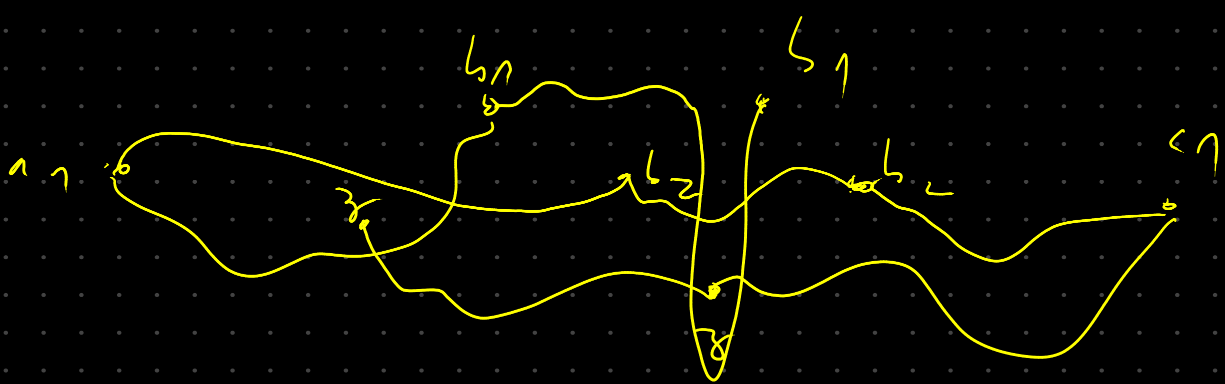
make sure $c_2$ element don't change the relation of $c_1$

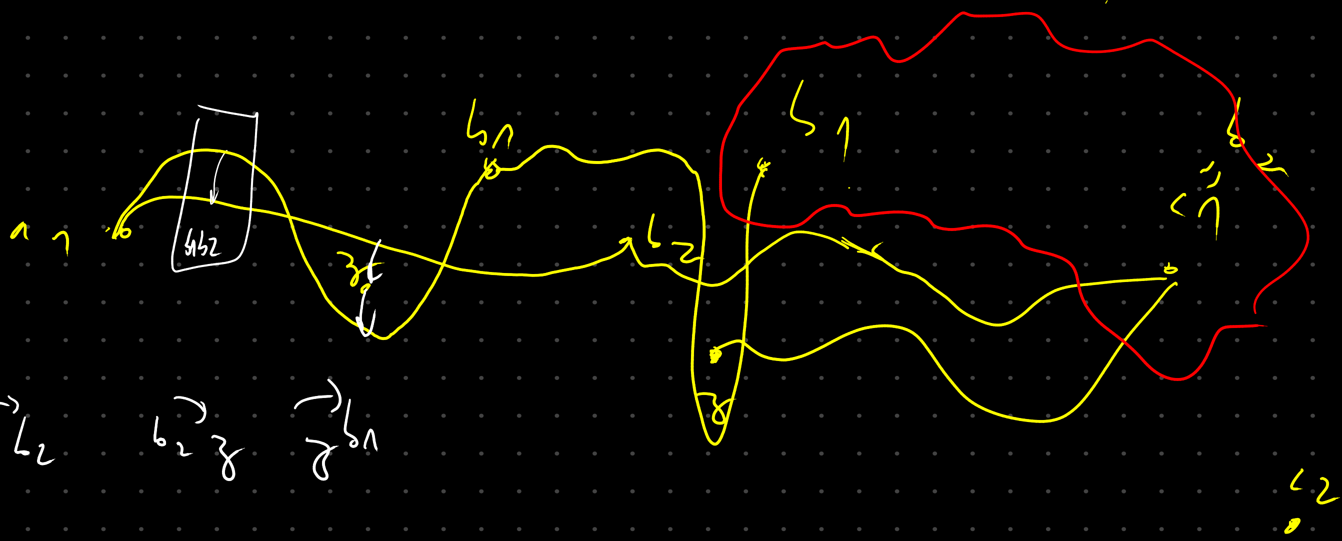
$b_2, b_7, b_4 \in c_1$

$$\begin{tikzcd} b_2 \arrow[r] \arrow[rr, bend right] & b_1 \arrow[r] & b_3 \end{tikzcd}$$



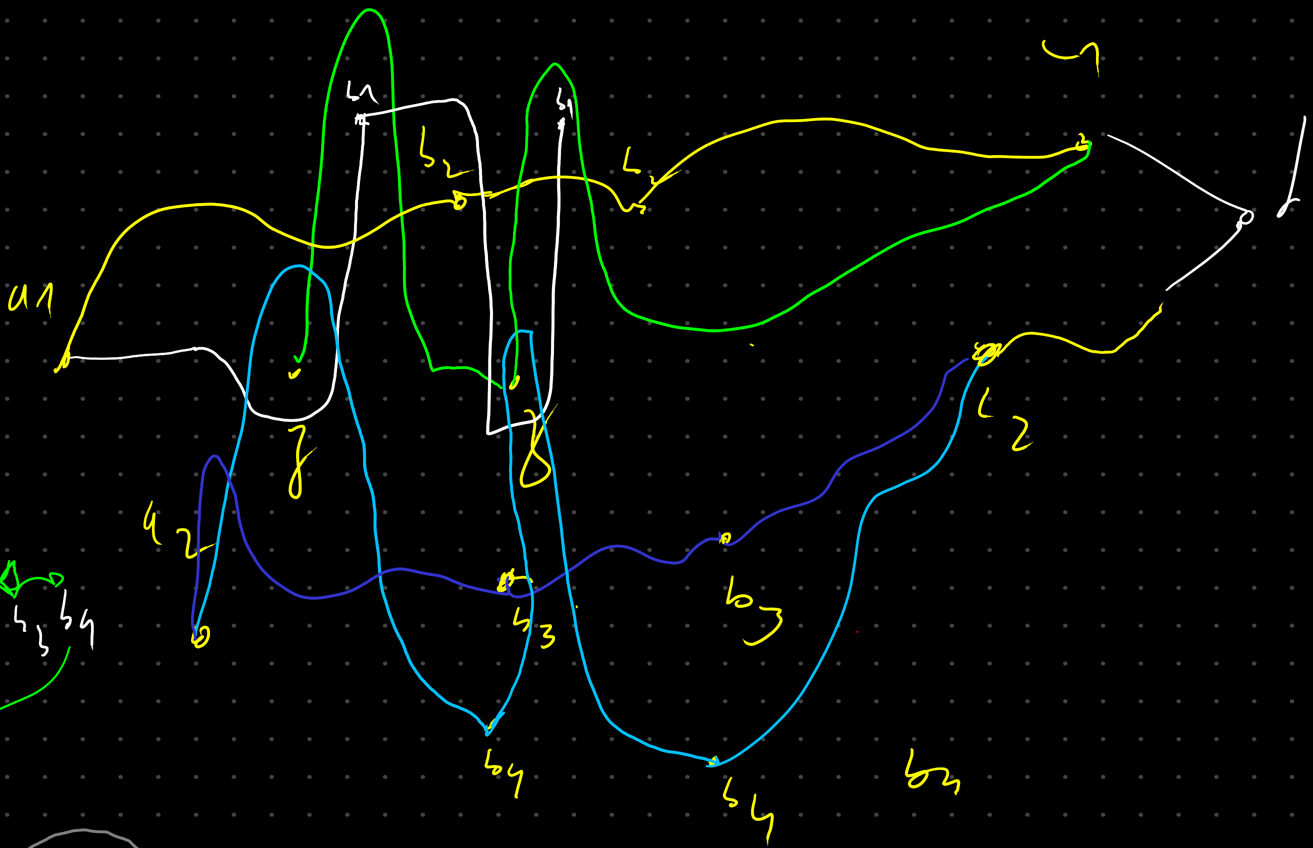


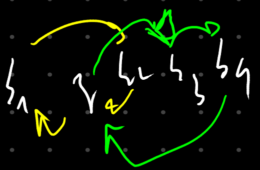


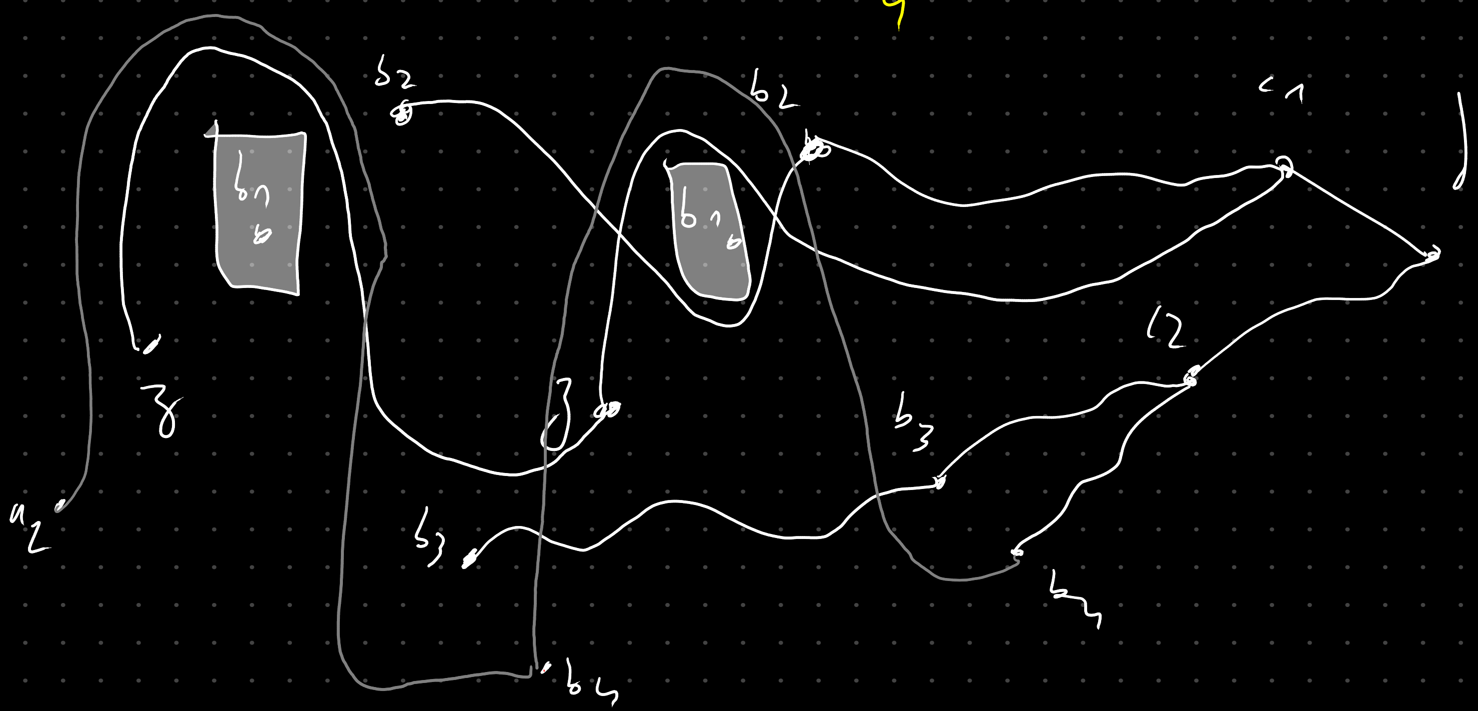


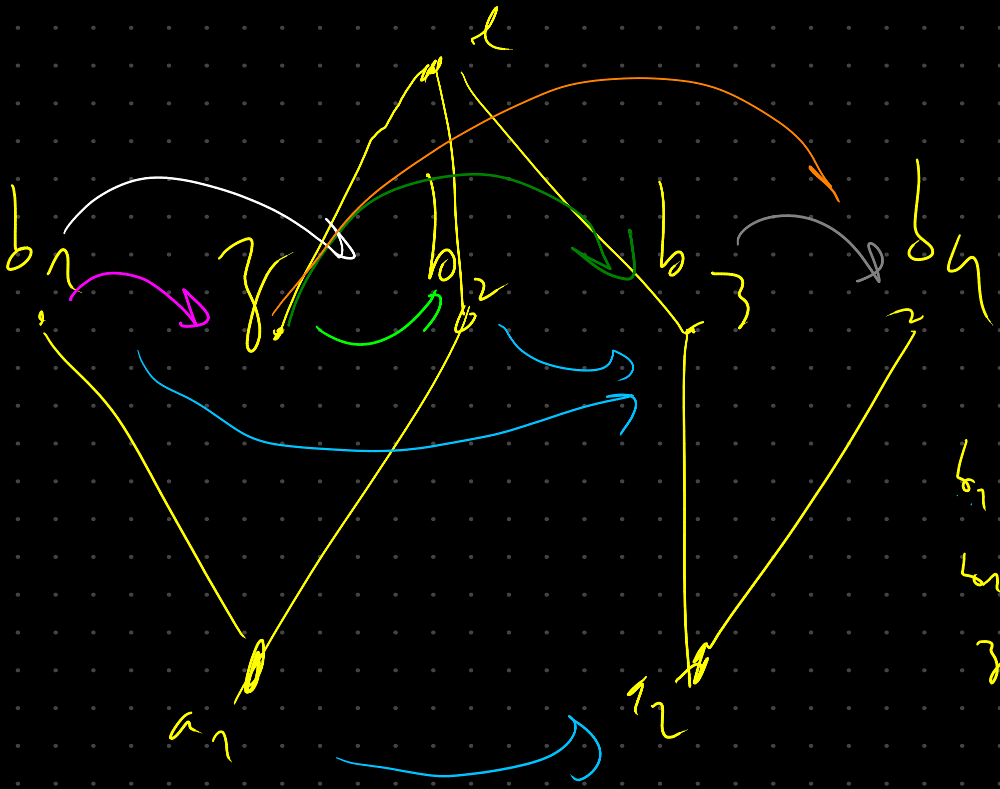

$\overrightarrow{b_1 b_2}$ $\overrightarrow{b_2 z}$ $\overrightarrow{z b_1}$

6





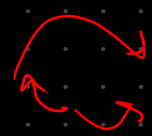

$b_1 b_3$
$b_1 b_4$
$b_2 b_3$
$b_2 b_4$
$a_1 a_2$

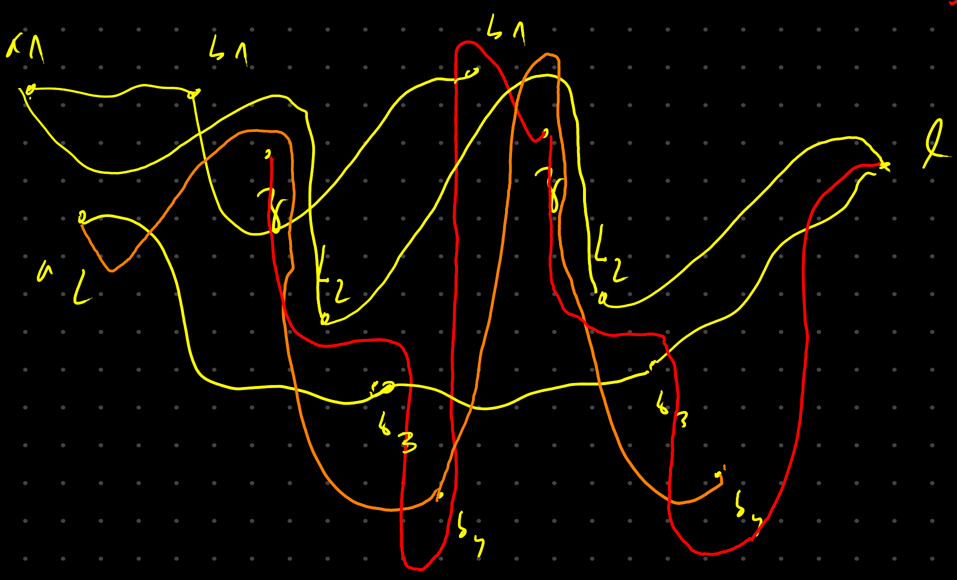
$b_1 b_2 = 1$ $\quad$ $b_3 b_4 = 1$

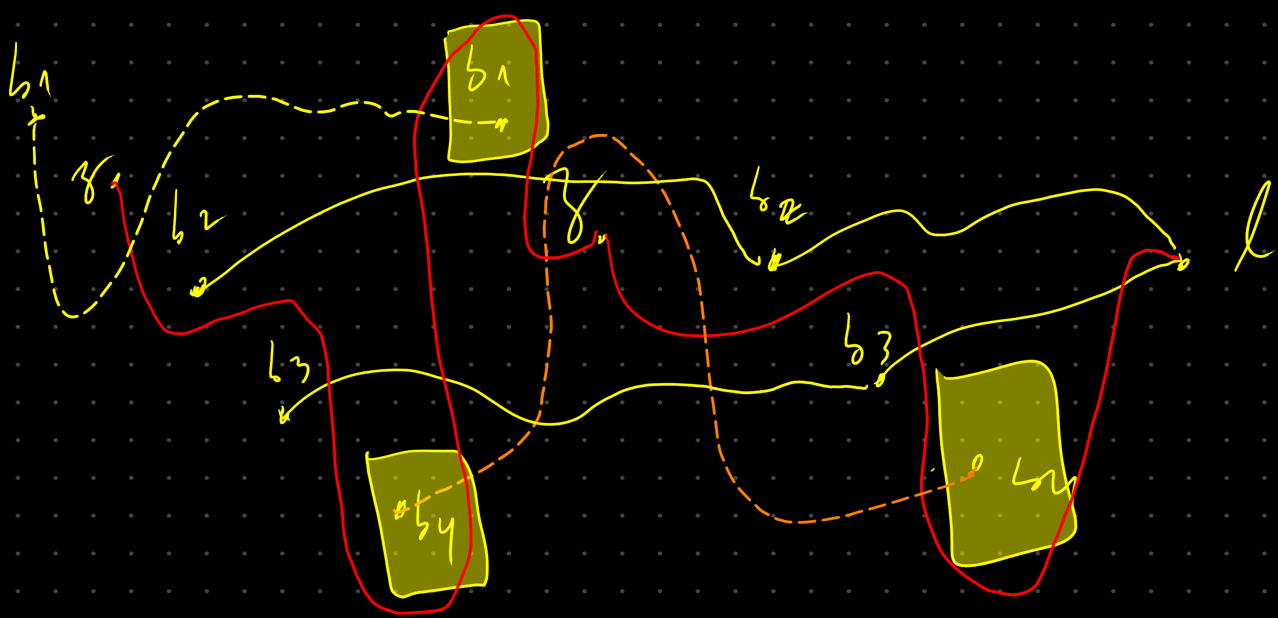
$b_1 z = 0$ $\quad$ $b_3 z = 0$

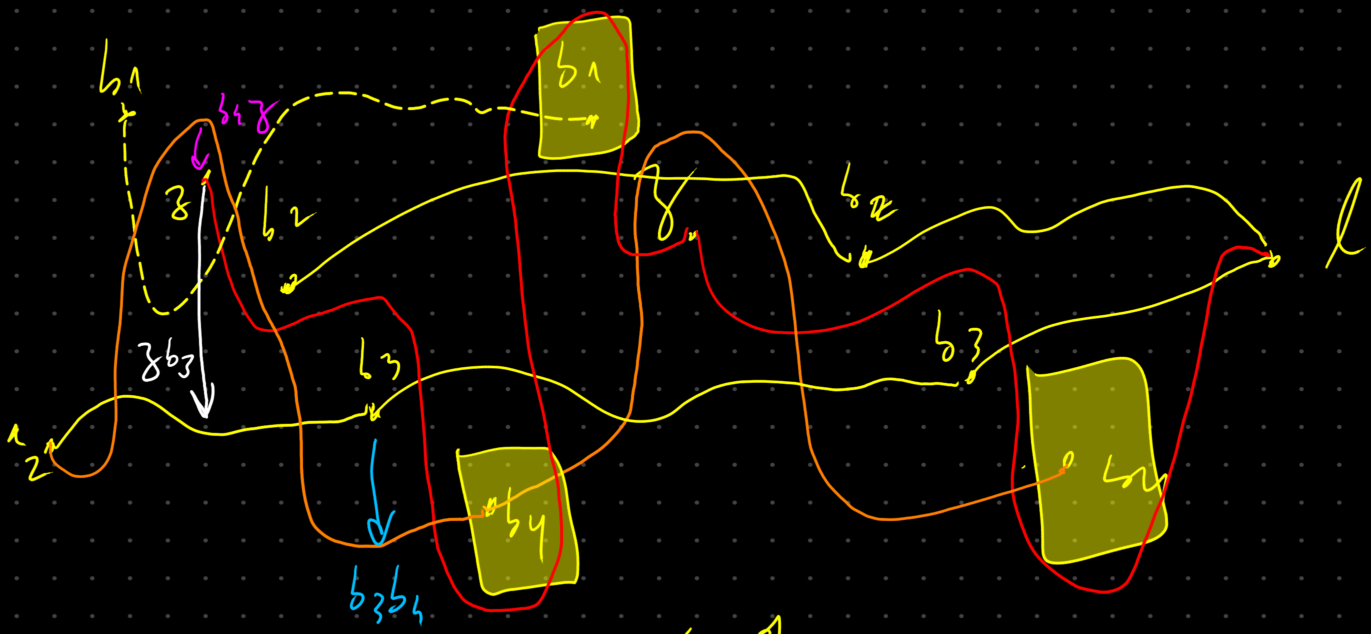
$z b_2 = 0$ $\quad$ $z b_4 = 0$

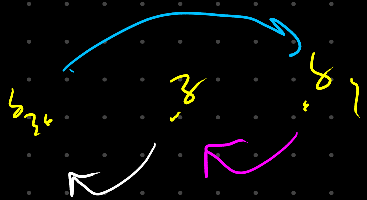
$z b_3 = 1$



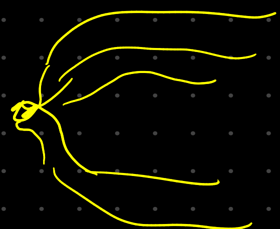


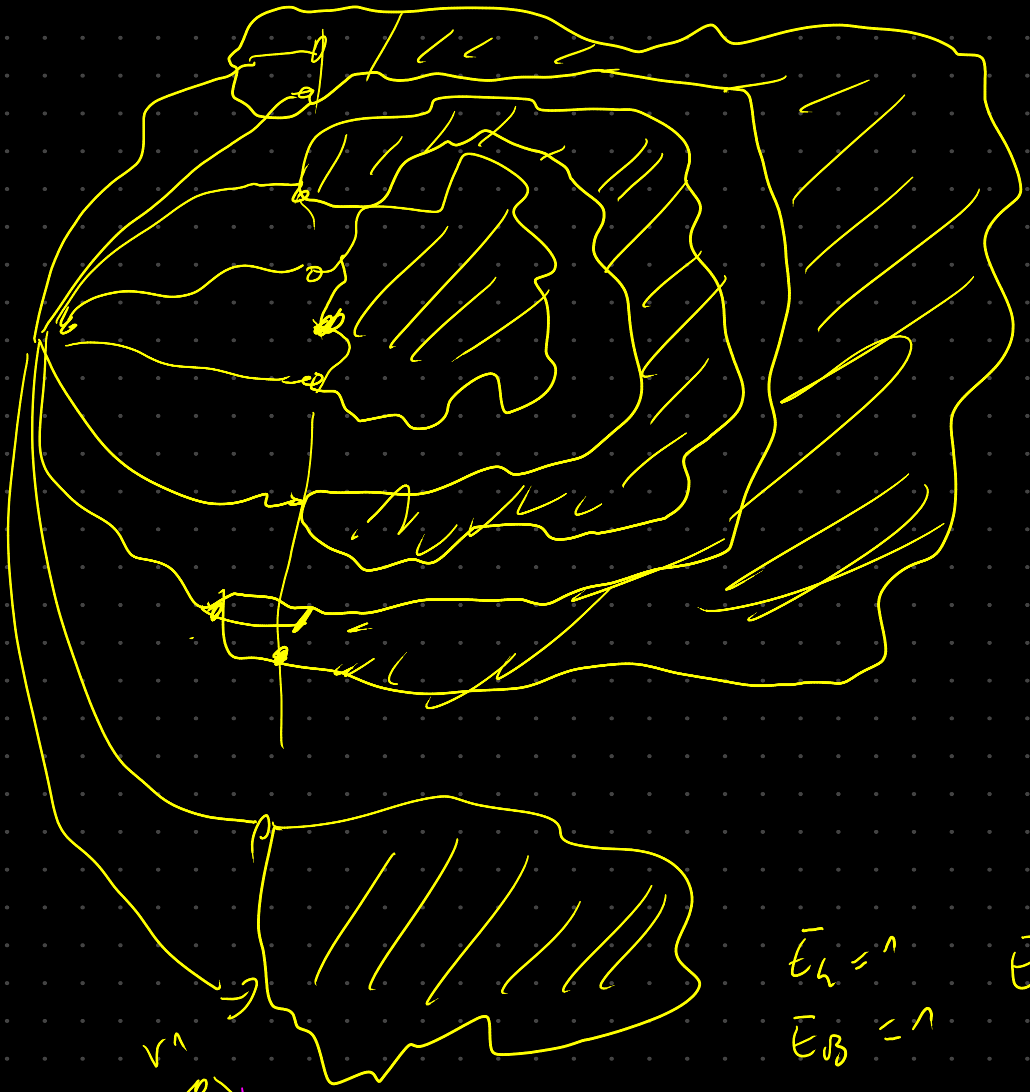



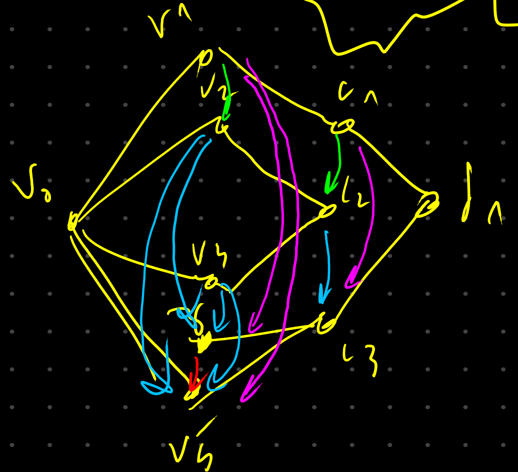


c of
the V vertex that is not
connected to z is
smaller than the
cycle of the c1





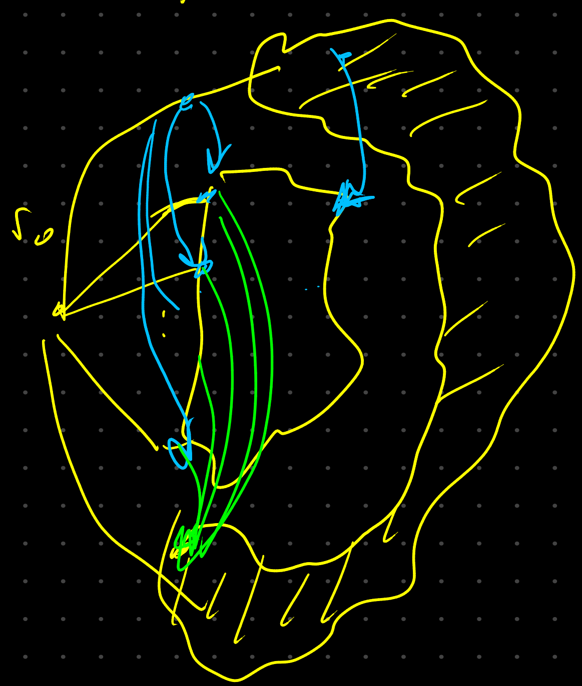


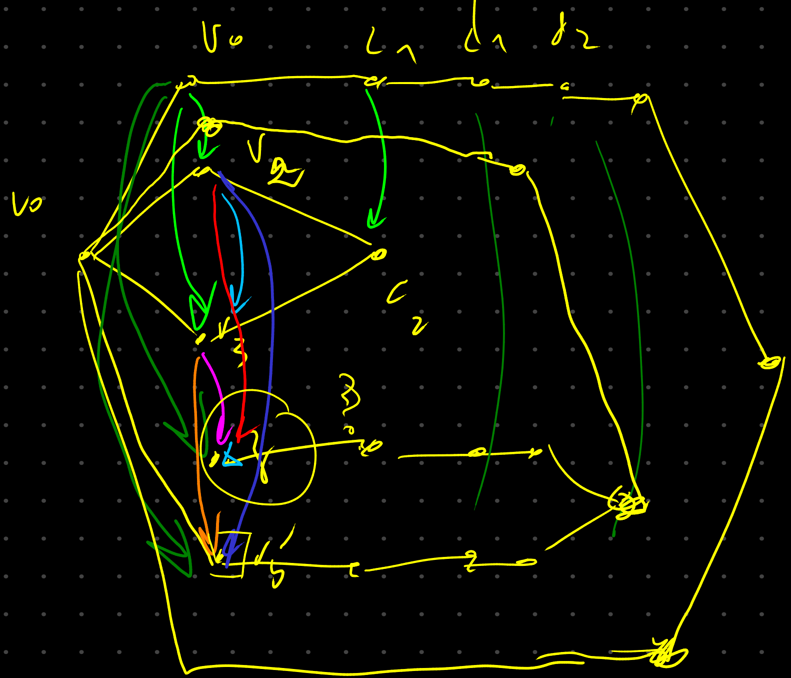

$\bar{E}_L = 1$ $\bar{E}_M = 1$

$\bar{E}_B = 1$

if we put $E_R = 1$

| $\bar{E}_L$ | $E_B$ | $\bar{E}_M$ | $\bar{E}_R$ |
|---|---|---|---|
| 1 | 1 | 1 | X |

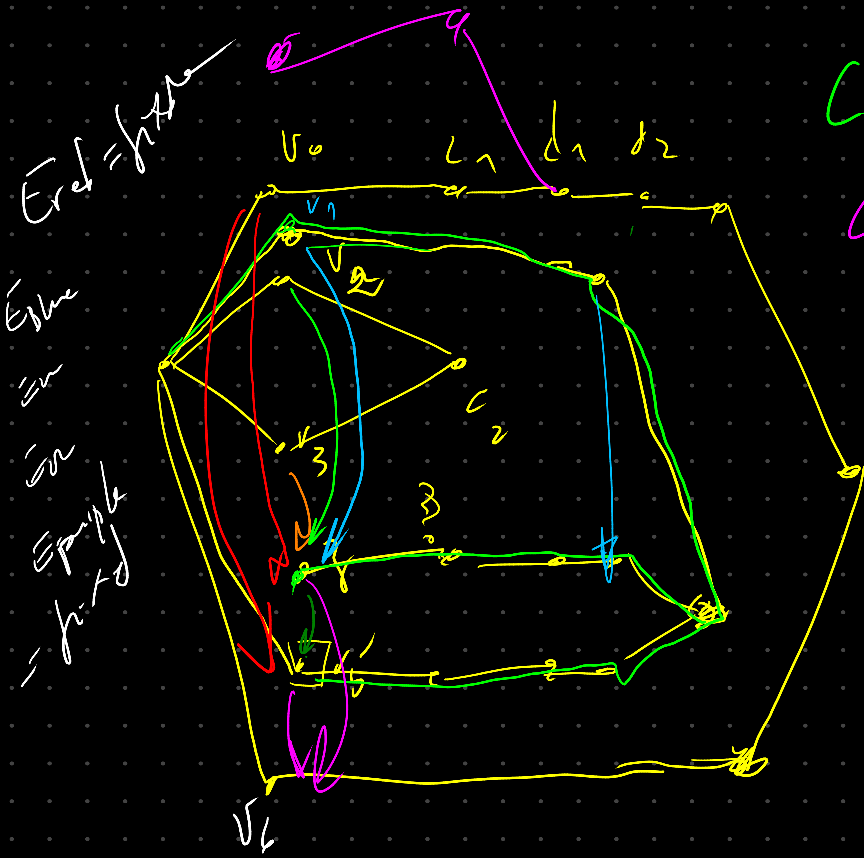



We take first fix the order of the vertices in the maximal component, we fix after the order of the closest smaller comp by putting it in it's order in the rotation (we determined the rotation before hand)

Meaning that the order btw. endpoints of adjacent edges stays the same, the ch

so the hanging vertex has the same order as its connected comp.



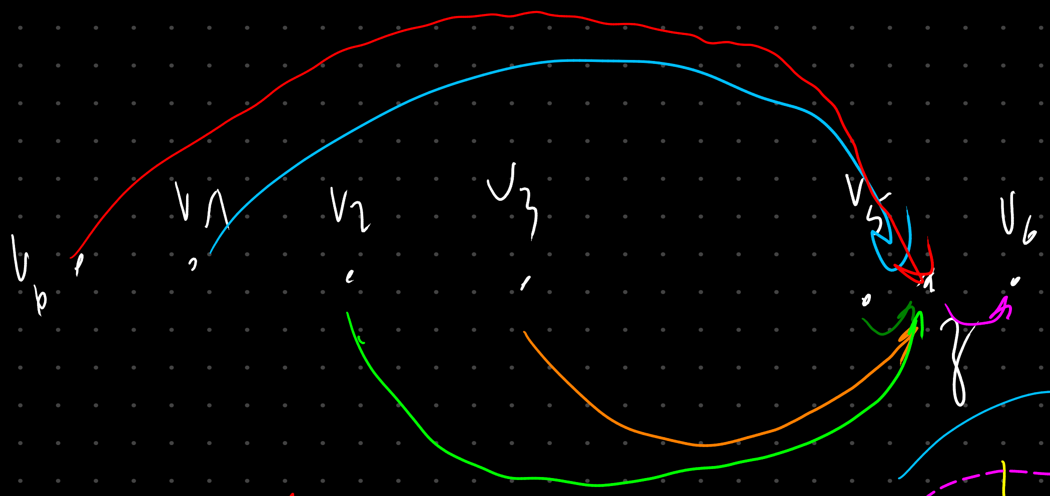

$C_{green} = \{v_4, v_5\}$

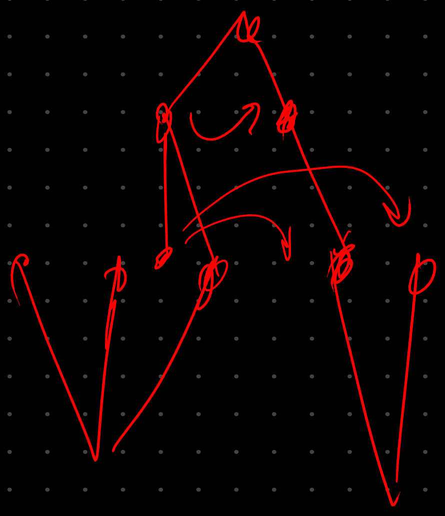
$C_n = \{v_2, v_3\}$

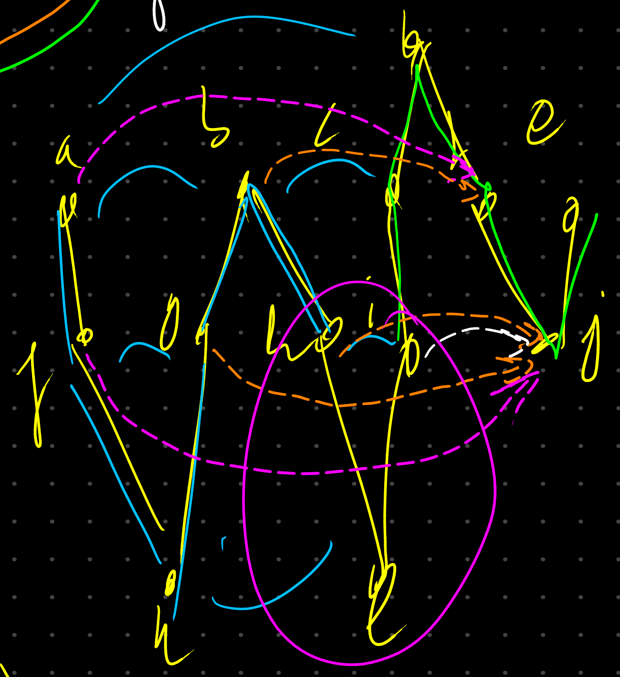
$c_g > C_{mag}$

$\{ v_1\gamma, \cancel{v_4\gamma}\ v_5'\gamma$

$\gamma v_6 = v_5 v_6 \qquad v_0\gamma = v_0 v_5 \qquad v_1\gamma,\ v_2\gamma,\ v_3''\gamma$





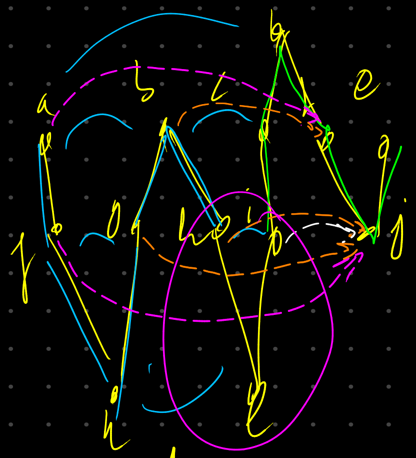


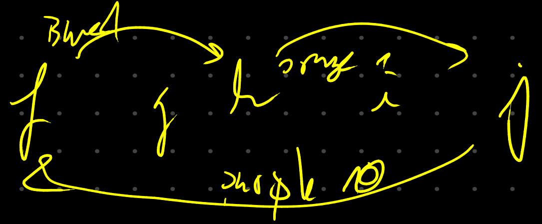
fj , (gj, hj) , (ij)

1 1 1

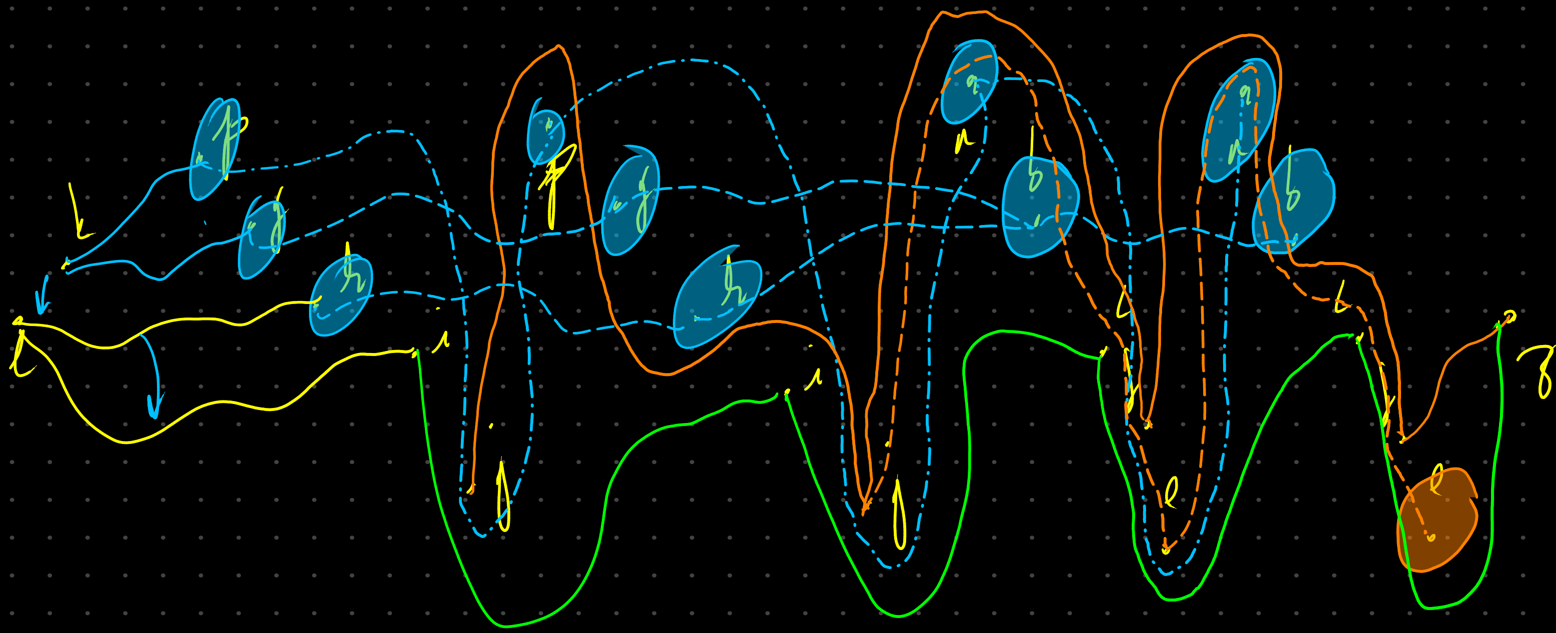
0 0 0

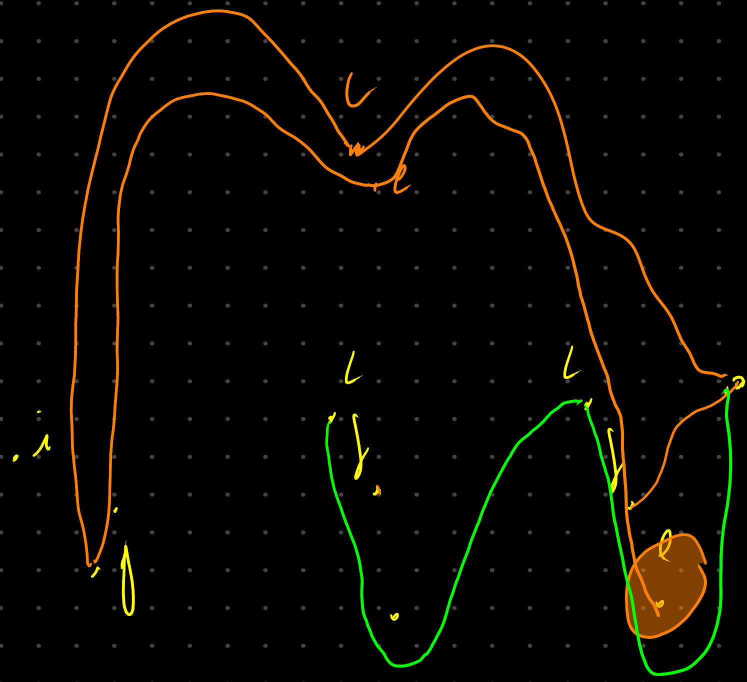
we can think of this vertex as the last one drawn in it's level (the order of inter-level don't matter)

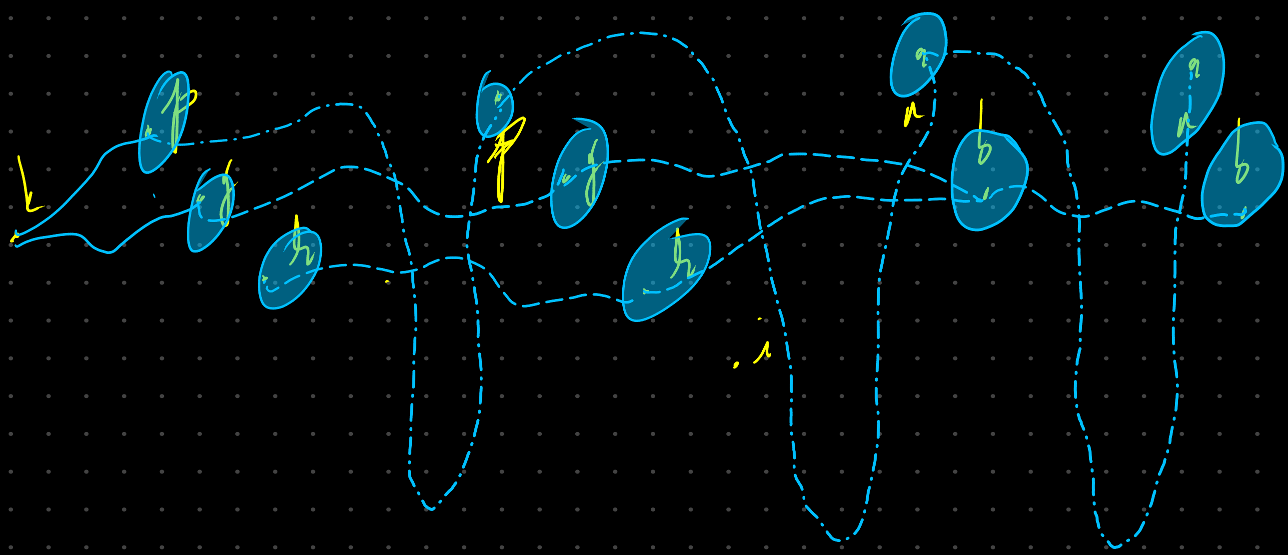


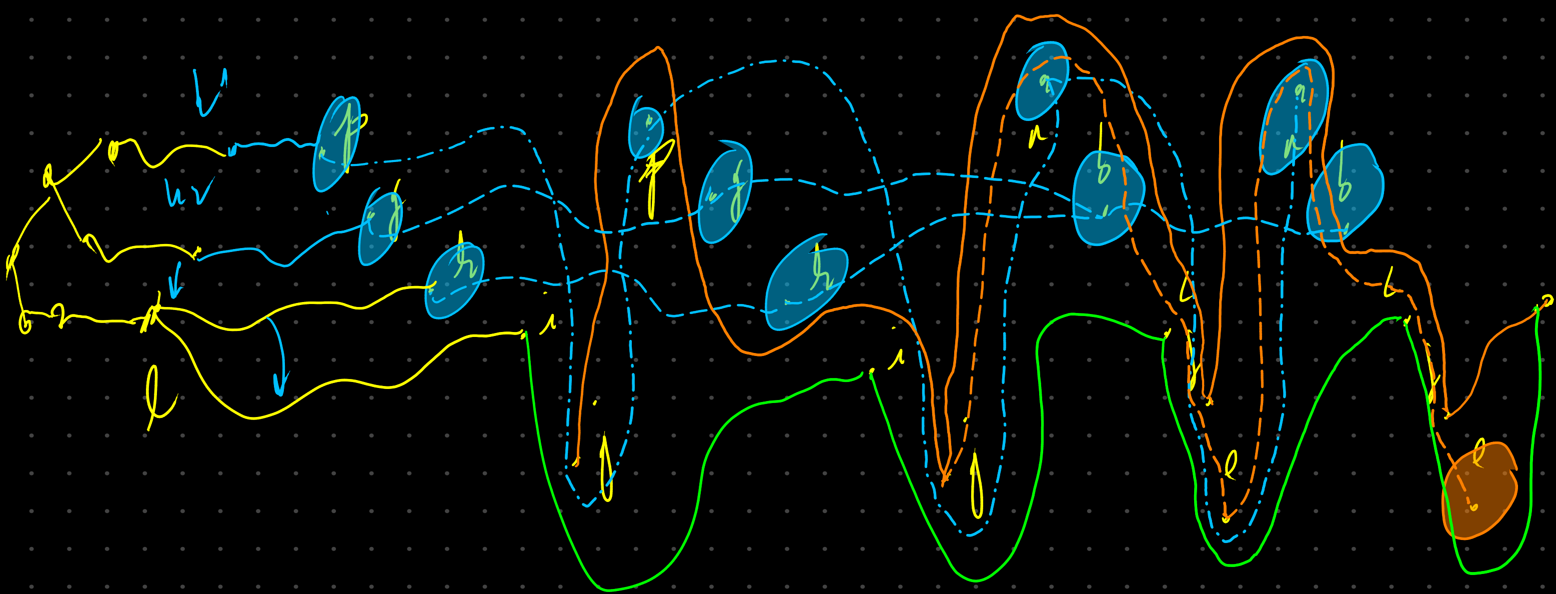


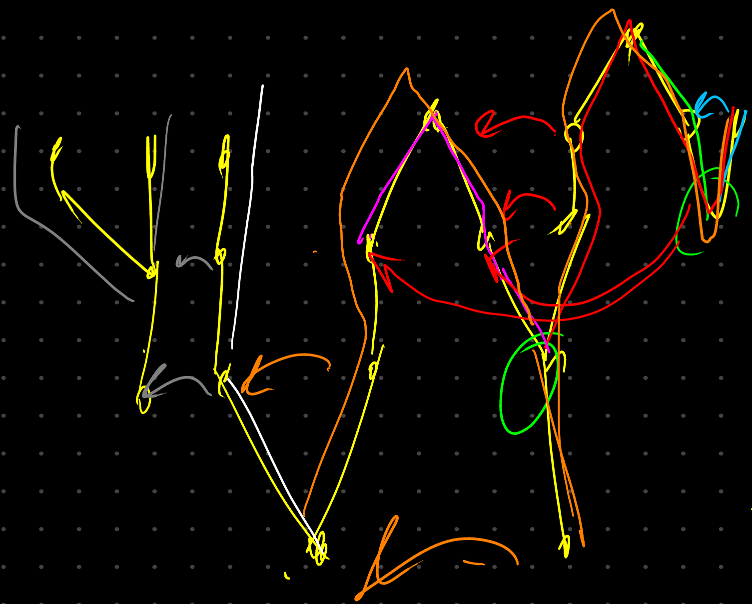
gh,
hi
jf











it helps us reach that vertices we can't reach in $\neg$ eq class ~~by~~ that we already had

we draw the graph from $k$ and pick a direction left or right n visiting

we start reconstructing our graph

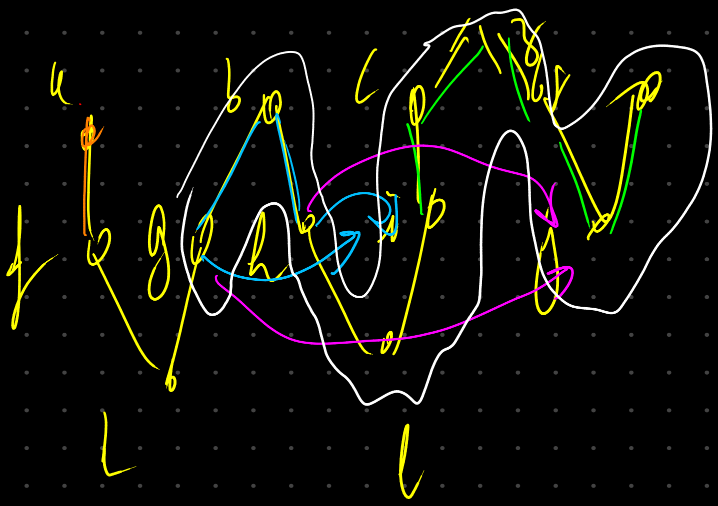
when we add a vertex that is adjacent to multiple edges

if $k - \{v\}$ gives us multiple comps.

we fix $eq(l_1, l_2)$

$$eq(l_1, l_2) = \forall u \in l_1 \ \forall v \in l_2 \ Eq(u, v) =$$

$$\bigwedge_{\substack{u \in l_1 \\ v \in l_2}} Eq(u', v')$$

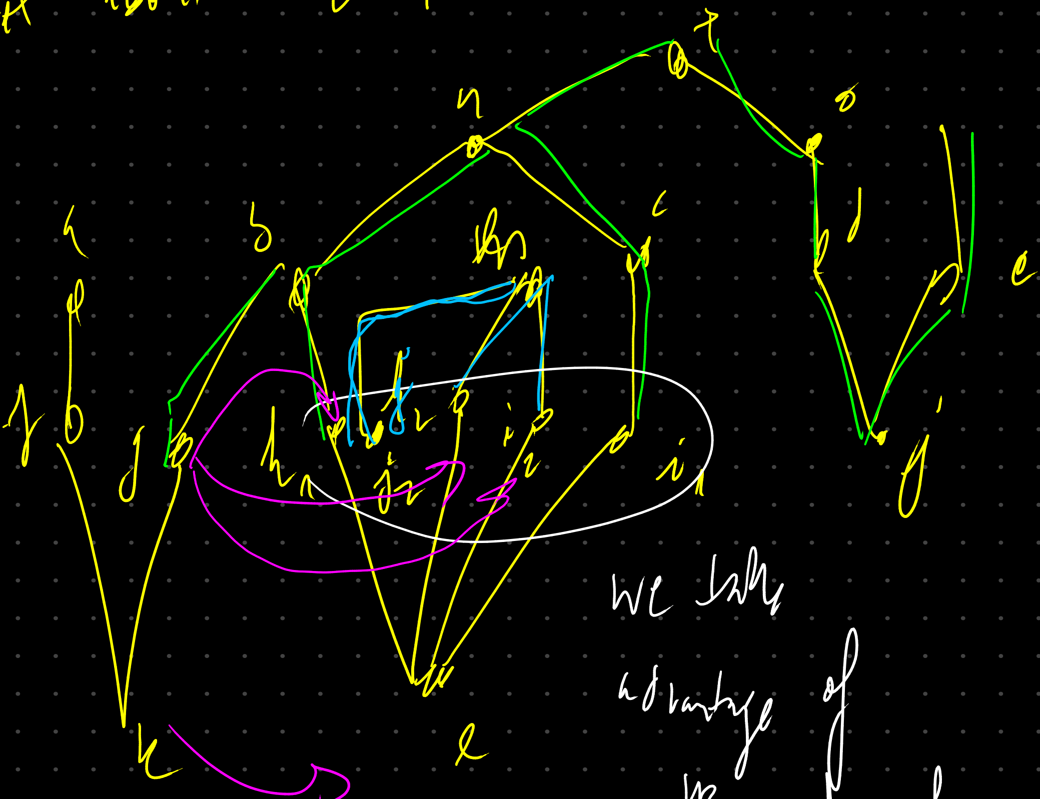


$S = \{\{gi, gj, hi, hj\}\}$

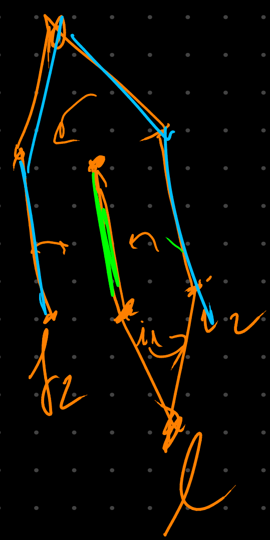
$Eq_b = Eq_n = Eq_r$

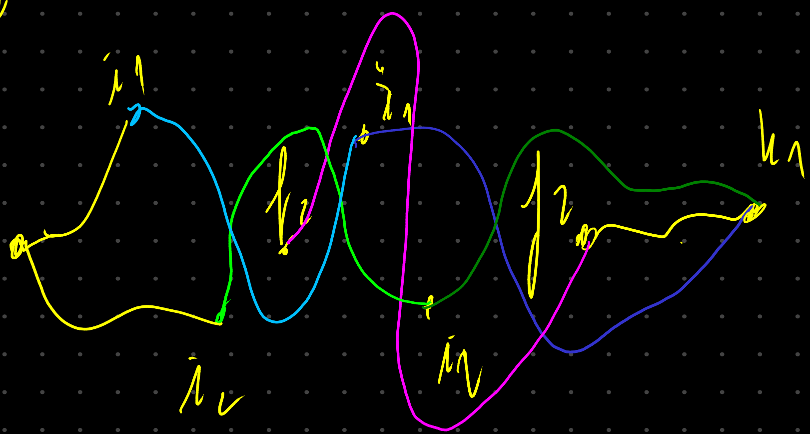
$fg, fh, fi, fj$

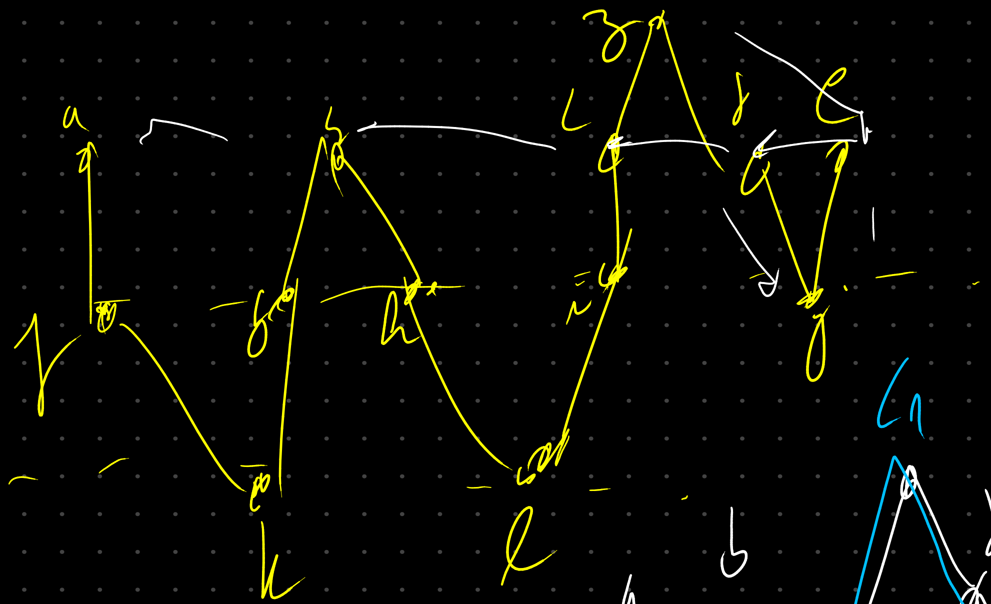
What about L comps inside others

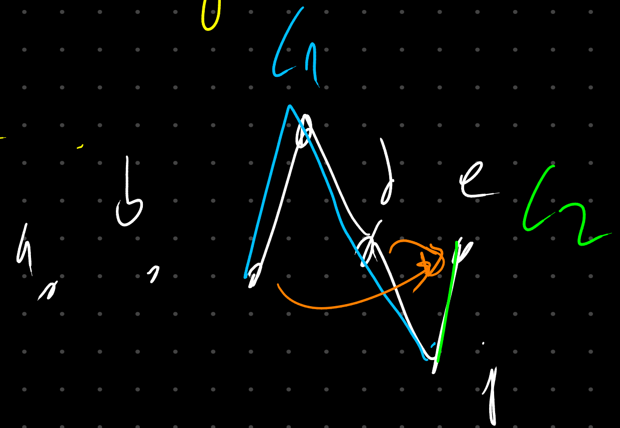


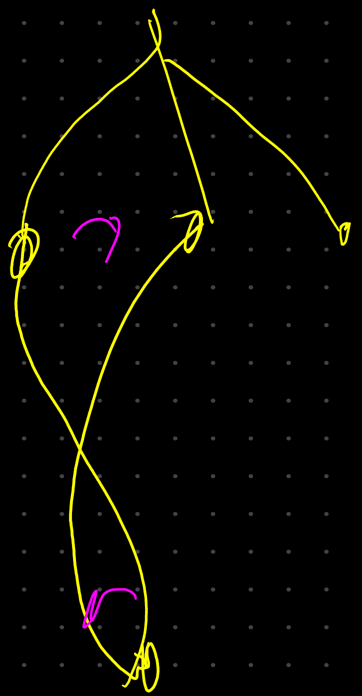
we take advantage of the order of adjacent edges

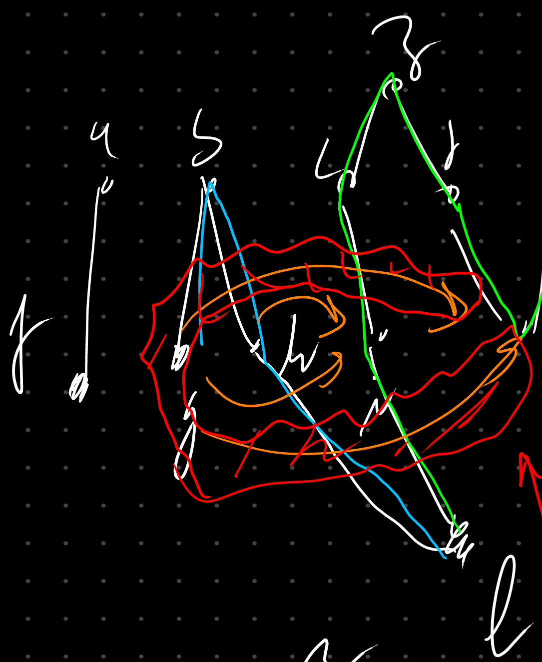




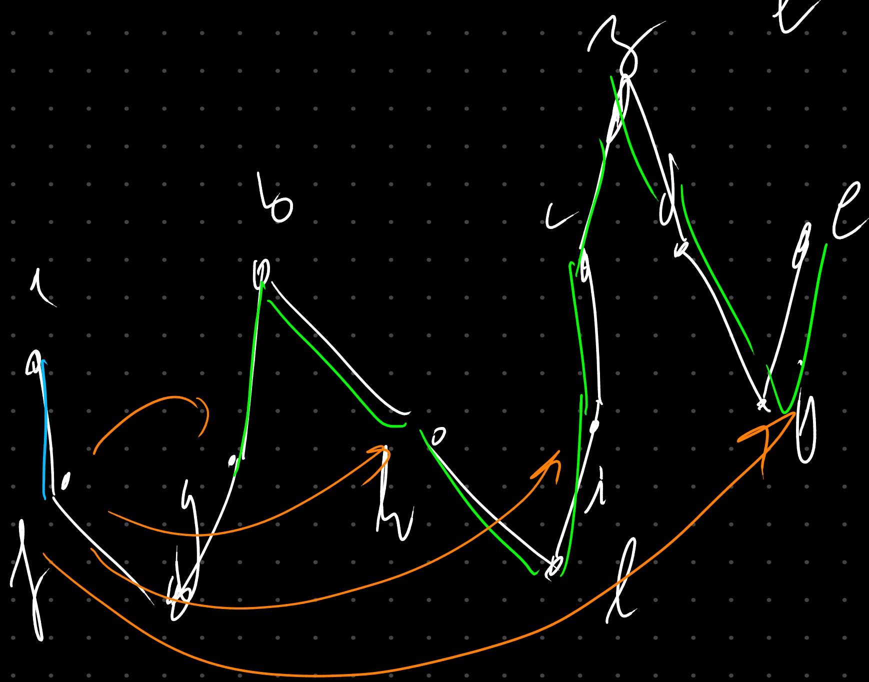








purpose of this



prove of correctness

$$S\ H\text{-}T \xrightarrow[\text{edges order}]{\text{fix adjacent}} W\ H\text{-}T$$

we don't change equivalent classes

$\downarrow$

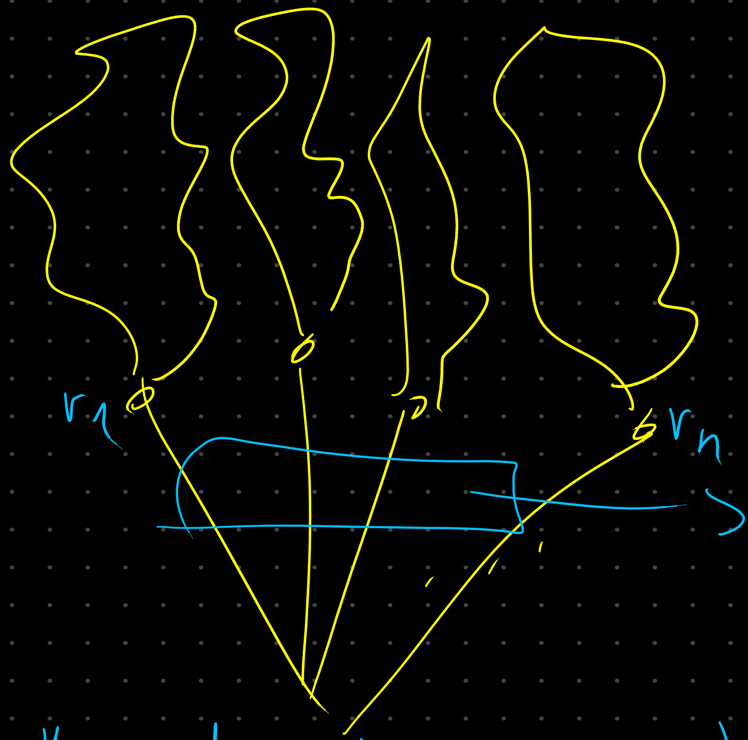
planarity ✓

1

fix trees

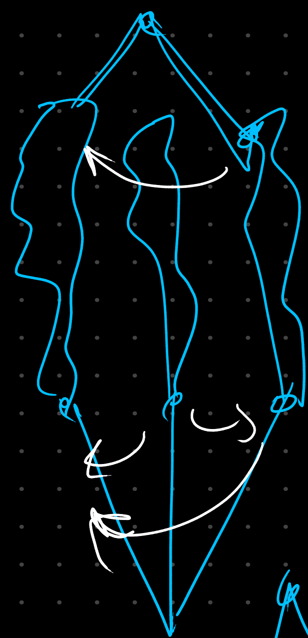
does it work for this too?

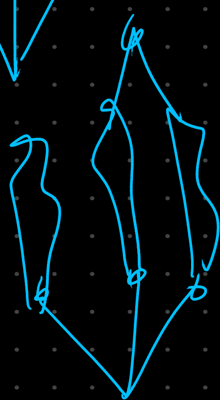
show that the sol. work for all of them

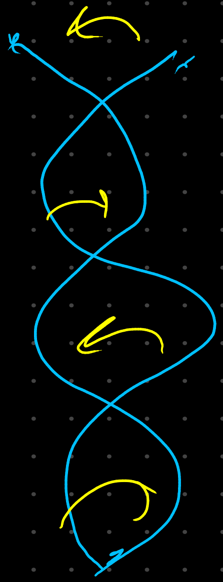
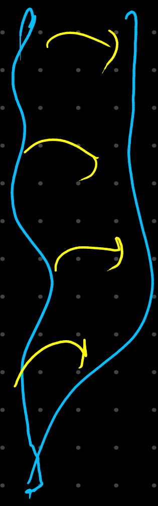
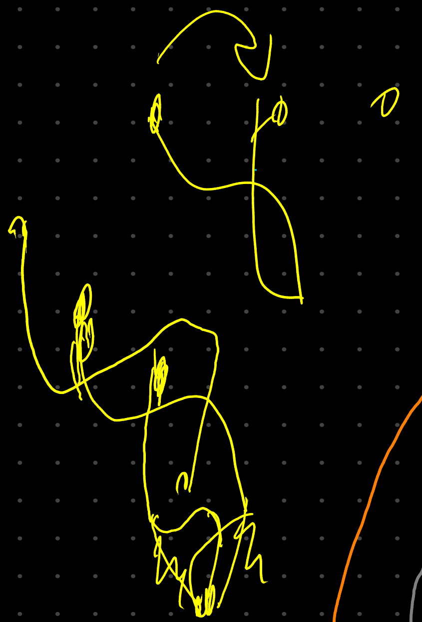
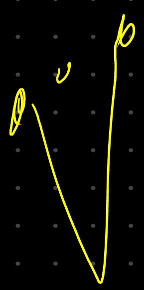
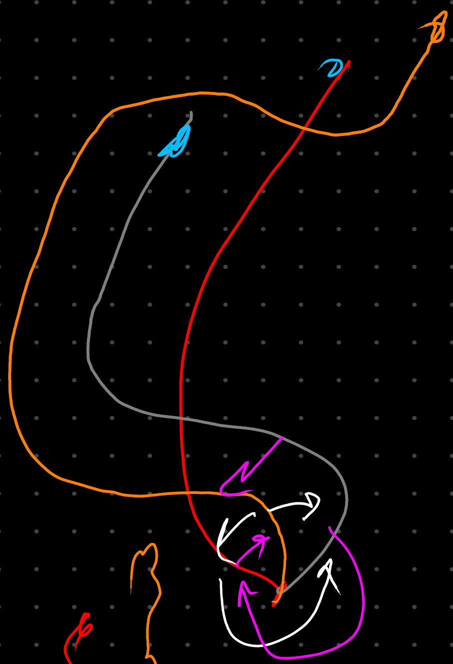
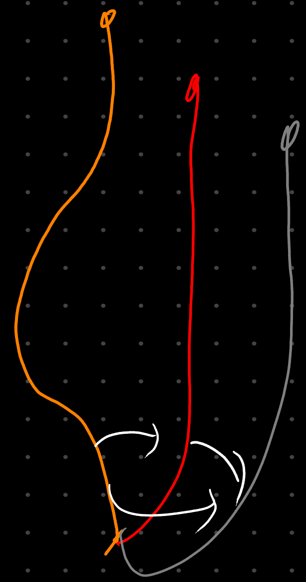



if there is any freedom here we want to make one that the $L_i$ dont tangle each other

the order of $(v_1, \dots, v_n)$ is the same of



$V_1 V_2$

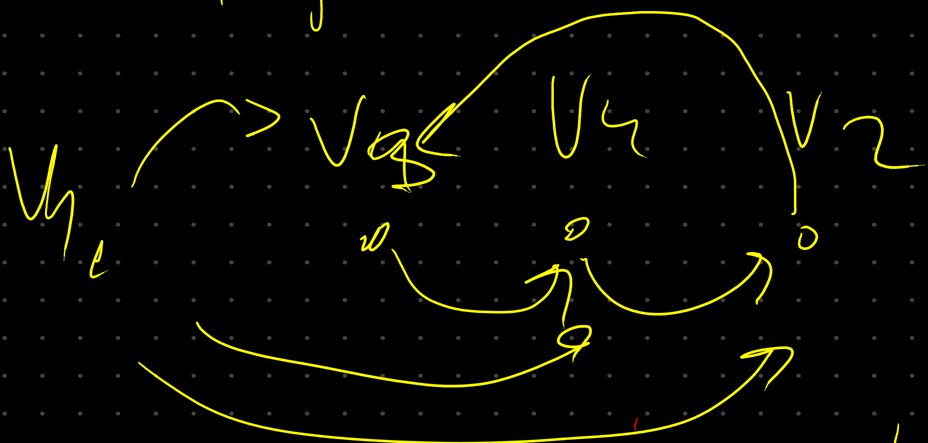
$V_1 V_2 V_3$

$V_1 V_2,\ V_1 V_3,\ V_1 V_4$

$V_2 V_3,\ V_4 V_3$

$V_1 V_2,\ V_1 V_3,\ V_1 V_4,\ V_2 V_3,\ V_4 V_2$

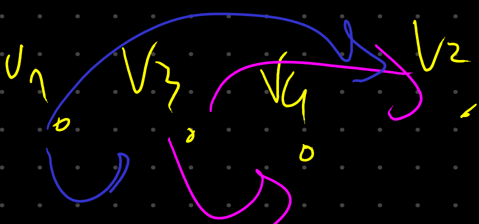
$V_3 V_4$



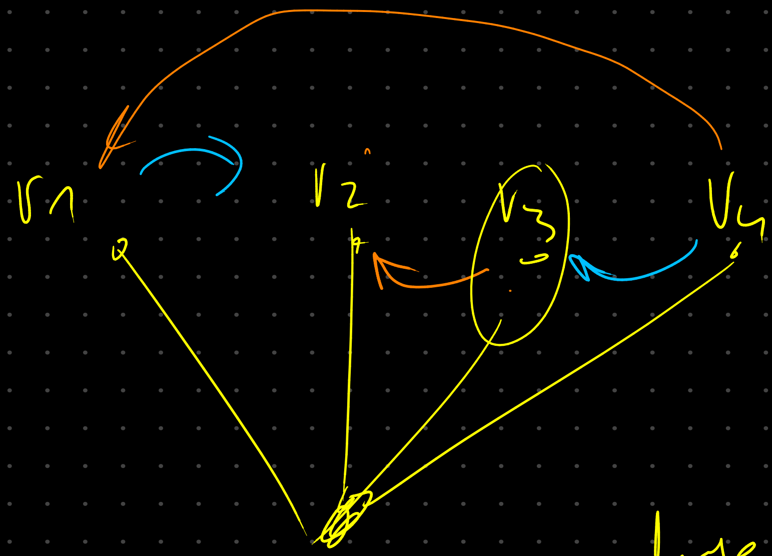
$Eq(V_1 V_2) = \{V_1 V_2,\ V_1 V_2\}$

$V_1 V_4\ (V_2 V_3) = \{V_1 V_3,\ V_2 V_3\}$

$V_2 V_4$



$V_1 V_2,\ V_1 V_3,\ V_1 V_4,\ V_2 V_3,\ V_4 V_3,\ V_2 V_4$




$V_1 V_2$ — $V_4 V_3$

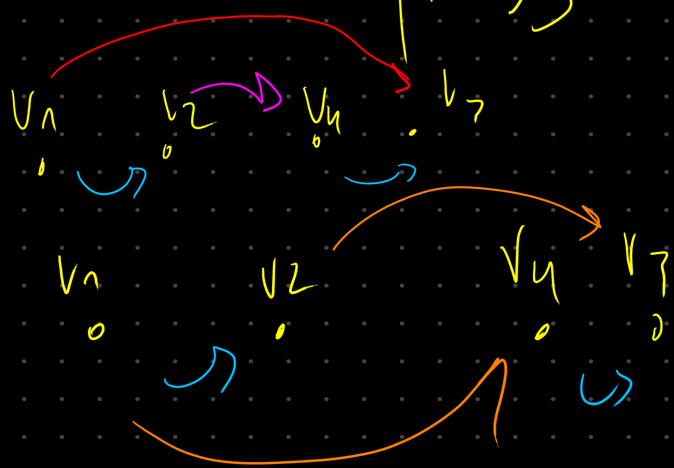
$V_1 V_3$ — $V_3 V_2$

| | |
|---|---|
| $V_1 V_4$ | $V_4 1$ |
| $V_2 V_4$ | $V_4 2$ |

$V_1 V_2$ — $V_4 V_3$

we dump this

$V_1 V_3$




we put one eq class if we need to dump the the order for the eq to point to the same direction we do it.

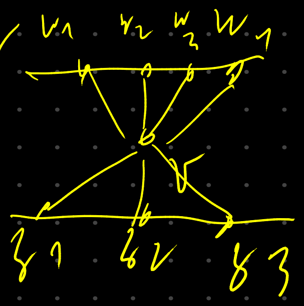
We put the relation as the order and fix the order by swapping the others will invert direction of the same eq.

Impl:

Big tables of the Eq

By level $h$

$v \mid \ell(v) = h$




→ upper side $h-1$

→ lower side $h+1$

↓ reorder based on Eq

$G' = G' \sqcup \{v\}$

cut vertex in $G'$?

↳ yes →


order the $Eq(C_i, C_j)$

$\forall i,j \in \{1,..,3\}$ $i<j$

framework of HDF helper functions:

ogdf::connectedComponents

ogdf::findCutvertices